

In Memory computing Accelerator :

Problem :

In traditional disk based processing, we often face issues when computing large volumes of data. This is because the entire processing is done by the processor and data needs to be moved from the memory to the processor and back to the memory , while the memory has no active role and only stores the data that needs to be processed. When large amounts of data are involved, significant latency is observed because the data has to travel repeatedly from the memory to the processing unit. This data movement creates a major bottleneck and increases the overall processing time.

Memory to processor to memory .

Design a Processing In Memory (PIM) module that performs matrix-vector multiplication directly using analog computation in a resistive crossbar array ,eliminating data movement bottleneck . Achieve 10x energy efficiency vs. traditional MAC architectures (multiply accumulate). _